SINDH MADRESSATUL ISLAM UNIVERSITY KARACHI

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LAB 01

ADVANCE STATISTICS

LAB INSTRUCTOR: MISS AQSA

Course: Advance Statistics

Département: Data Science

Semester:4

Section: A

Advance Statistics Lab 4

Objective:

To understand and calculate the relationship between two variables using Correlation and Simple Linear Regression in Microsoft Excel.

Theory:

1. Correlation
 Measures how strongly two variables are related.
 Value of r ranges from -1 to +1:
  +1: Perfect positive relationship
  -1: Perfect negative relationship
  0: No relationship
- Excel Formula: =CORREL(array1, array2)

2. Regression
- Regression shows how one variable can predict another.
- Equation: $Y = a + bX$
  Y: Dependent variable
  X: Independent variable
  a: Intercept
  b: Slope
- In Excel, regression is performed using the Data Analysis.

Example Dataset :

Student	Study Hours (X)	Marks (Y)
A	2	30
B	4	45
C	6	55
D	8	65
E	10	80

Steps in Excel:

Step 1: Enter Data
Type the above dataset in Excel columns A1:C6.

Step 2: Find Correlation
Use the formula =CORREL(B2:B6, C2:C6)

Step 3: Run Regression using Data Analysis Toolpak
1. Go to Data → Data Analysis → Regression
2. Input Y Range: C1:C6 (Marks)
   Input X Range: B1:B6 (Hours)
3. Check Labels and Output results to a new worksheet
4. Click OK

Step 4: Create a Scatter Plot
Select both columns → Insert → Chart → Scatter → Add Trendline (Display Equation and R^2)

Interpretation Example:

If the equation is Y = 20 + 6X, then for 9 study hours:

Y = 20 + (6 × 9) = 74 marks

Student Task:

Use the following dataset and perform Correlation and Regression in Excel.

Month	Advertisement Budget ($1000)	Sales ($1000)
1	5	25
2	7	30
3	9	35
4	11	45
5	13	50
6	15	55
7	17	60
8	19	65

Steps:
1. Enter the dataset in Excel.
2. Calculate Correlation using =CORREL(B2:B9, C2:C9).
3. Perform Regression using the Data Analysis Toolpak.
4. Record Regression equation and R Square value.
5. Create a Scatter Plot and add a Trendline with equation and R^2.
6. Predict Sales when Advertisement Budget = $20,000 using regression equation.

Questions:

1. What does the correlation value indicate about the relationship between budget and sales?

2. What is the meaning of R Square in your output?

3. What is the slope and intercept of your regression equation?

4. How much would you expect sales to increase if the budget increases by $1,000?